

PROBLEM-SOLVING EXERCISE

Can an insect outbreak threaten a forest's ability to absorb CO₂ from the atmosphere?

One way to combat climate change is to plant trees, since trees absorb large amounts of CO₂ from the atmosphere, converting it to biomass through photosynthesis. But what happens to the carbon stored as biomass in trees when an insect population explodes in number? Such insect outbreaks have become more frequent with climate change.



A tree with dozens of "pitch tubes," indications of a damaging outbreak of mountain pine beetles (inset).

In this exercise, you will test whether an outbreak of the mountain pine beetle (*Dendroctonus ponderosae*) alters the amount of CO₂ that a forest ecosystem absorbs from and releases to the atmosphere.

Your Approach The principle guiding your investigation is that every ecosystem both absorbs and releases CO₂. Net ecosystem production (NEP) indicates whether an ecosystem is a carbon sink (absorbing more CO₂ from the atmosphere than it releases; this occurs when NEP > 0) or a carbon source (releasing more CO₂ than it absorbs; NEP < 0). To find out if the mountain pine beetle affects NEP, you will determine a forest's NEP before and after a recent outbreak of this insect.

Your Data From 2000 to 2006, an outbreak of the mountain pine beetle killed millions of trees in British Columbia, Canada. The impact of such outbreaks on whether forests gain carbon (NEP > 0) or lose carbon (NEP < 0) was poorly understood. To find out, ecologists estimated net primary production (NPP) and cellular respiration by decomposers and other heterotrophs (R_h), before and after the outbreak. These data allow forest NEP to be calculated from the equation NEP = NPP – R_h.

	NPP [g/(m ² · yr)]	R _h [g/(m ² · yr)]	NEP [g/(m ² · yr)]
Before outbreak	440	408	
After outbreak	400	424	

Your Analysis

- Complete the table by calculating the values of NEP before and after the outbreak. Before the outbreak, was the forest a carbon sink or a carbon source? After the outbreak?
- NEP is often defined as NEP = GPP – R_T, where GPP is gross primary production and R_T equals cellular respiration by autotrophs (R_a) plus cellular respiration by heterotrophs (R_h). Use the relation NPP = GPP – R_a to show that the two equations for NEP introduced in this exercise are equivalent.
- Based on your results in question 1, predict whether the mountain pine beetle outbreak could have feedback effects on the global climate. Explain.

➔ **Instructors:** A version of this Problem-Solving Exercise can be assigned in **Mastering Biology**.

Climate change can also affect whether an ecosystem stores or loses carbon over time. As discussed earlier, net ecosystem production, or NEP, reflects the total biomass accumulation that occurs during a given period of time. When NEP > 0, the ecosystem gains more carbon than it loses; such ecosystems store carbon and are said to be a carbon *sink*. In contrast, when NEP < 0, the ecosystem loses more carbon than it gains; such ecosystems are a carbon *source*.

Recent research shows that climate change can cause an ecosystem to switch from a carbon sink to a carbon source. For example, in some arctic ecosystems, climate warming has increased the metabolic activities of soil microorganisms, causing an uptick in the amount of CO₂ produced in cellular respiration. In these ecosystems, the total amount of CO₂ produced in cellular respiration now exceeds what is absorbed in photosynthesis. As a result, these ecosystems—which once were carbon sinks—are now carbon sources. When this happens, an ecosystem may contribute to climate change

by releasing more CO₂ than it absorbs. Indeed, a 2017 study found that with climate warming, large regions of tundra in Alaska now release more CO₂ than they absorb—and for the years 2013 and 2014, the entire state of Alaska released more CO₂ than it absorbed. In the **Problem-Solving Exercise**, you can examine how outbreaks of an insect population may affect the NEP of forest ecosystems.

CONCEPT CHECK 55.2

- Why is only a small portion of the solar energy that strikes Earth's atmosphere stored by primary producers?
- How can ecologists experimentally determine the factor that limits primary production in an ecosystem?
- WHAT IF?** Suppose a forest was heavily burned by a wild-fire. Predict how NEP of this forest would change over time.
- MAKE CONNECTIONS** Explain how nitrogen and phosphorus, the nutrients that most often limit primary production, are necessary for the Calvin cycle to function in photosynthesis (see Concept 10.3).

For suggested answers, see Appendix A.